

OpenFOAM-v1706 + hyStrath/hy2Foam

Validated Reproduction and Usage Guide

Release: v1.0.0-validated

Validated on Ubuntu 20.04 / GCC 7.5.0 / OpenFOAM-v1706

Guide date: 14 September 2026

Purpose: enable another user to download, verify, reconstruct, configure, and smoke-test the same OpenFOAM-v1706 + hyStrath/hy2Foam environment and the non-, partial-, fully-, and super-catalytic wall cases.

Repository: <https://github.com/singhbh943/OpenFOAM-v1706-hy2Foam>

Release: <https://github.com/singhbh943/OpenFOAM-v1706-hy2Foam/releases/tag/v1.0.0-validated>

Contents

- 1. Scope and validated baseline
- 2. Download and integrity verification
- 3. Reconstruct and extract the archive
- 4. Recommended runtime environment
- 5. OpenFOAM-v1706 and hyStrath environment setup
- 6. Confirm the hy2Foam executable and catalytic libraries

- 7. Catalytic-wall validation cases
- 8. Expected smoke-test results
- 9. Super-catalytic equilibrium verification
- 10. Troubleshooting and known pitfalls
- 11. Reproducibility checklist

1. Scope and validated baseline

This release is a preserved snapshot of a working OpenFOAM-v1706 + hyStrath/hy2Foam environment together with the associated catalytic-wall implementations and run/test cases. The validation described here is a controlled smoke/runtime validation, not a claim of fully converged aerodynamic or catalytic physics for every included case.

Component	Validated state
Operating environment	Ubuntu 20.04 container
Compiler	GCC/G++ 7.5.0
OpenFOAM	OpenFOAM-v1706
hyStrath	Fleming-era hyStrath CFD module
Solver	hy2Foam
User build option	linux64GccDPInt32Opt
Catalytic libraries	Partial, Fully, Super catalytic wall libraries
Production mesh	5-degree wedge mesh; checkMesh: Mesh OK
Smoke interval	0 to 1e-08 s

The archive also contains run directories and historical material. Some included example cases are not production-quality meshes. Follow the validated case paths and warnings in this guide.

2. Download and integrity verification

The complete environment is stored in GitHub Releases as a split gzip-compressed tar archive. Download all three assets from release v1.0.0-validated:

- OpenFOAM-v1706-hy2Foam-validated-20260914.tar.gz.part-00
- OpenFOAM-v1706-hy2Foam-validated-20260914.tar.gz.part-01
- OpenFOAM-v1706-hy2Foam-validated-20260914.SHA256SUMS

Expected SHA-256 values:

```
7129bf024e57f47a3c32dec29dd4d37eccc19a11e2fa5251663ca00bc404f81c  OpenFOAM-v1706-hy2Foam-validated-20260914.tar.gz.part-00
2860ba19c2de9ca1f61cd4e549503d98b53ecd50a8a08a81fcb0de89d6a9bfe4  OpenFOAM-v1706-hy2Foam-validated-20260914.tar.gz.part-01
```

Verify the downloaded parts:

```
sha256sum -c OpenFOAM-v1706-hy2Foam-validated-20260914.SHA256SUMS
```

Both archive parts must report OK before reconstruction. A checksum mismatch means the file should be downloaded again rather than used.

3. Reconstruct and extract the archive

```
cat OpenFOAM-v1706-hy2Foam-validated-20260914.tar.gz.part-* \
> OpenFOAM-v1706-hy2Foam-validated-20260914.tar.gz

gzip -t OpenFOAM-v1706-hy2Foam-validated-20260914.tar.gz
echo "GZIP_CHECK_EXIT_CODE=$?"

tar -xzf OpenFOAM-v1706-hy2Foam-validated-20260914.tar.gz
```

The gzip check should return exit code 0. Extraction creates the top-level trees OpenFOAM-v1706, ThirdParty-v1706, and bhimsingh-v1706.

4. Recommended runtime environment

The validated build was produced in an Ubuntu 20.04 container with GCC/G++ 7.5.0. Older OpenFOAM-v1706 and hyStrath source is sensitive to modern compiler changes, so an equivalent compiler/runtime environment is strongly recommended.

Item	Recommended / validated
Linux distribution	Ubuntu 20.04
GCC	7.5.0
G++	7.5.0
OpenFOAM arch	LSB; label=32; scalar=64
Floating point build	DP
Integer labels	Int32

The local Docker image used during validation was named `hystrath-of1706-gcc7`. That image name is an implementation detail of the validation machine and is not embedded as a portable Docker image in the GitHub release. Users should reproduce an equivalent Ubuntu 20.04/GCC7 environment if that image is not available.

5. OpenFOAM-v1706 and hyStrath environment setup

Assuming the extracted trees are placed under /home/<USER>/OpenFOAM, configure the shell as follows. Replace <USER> if your login name is different.

```
set +u
source /home/<USER>/OpenFOAM/OpenFOAM-v1706/etc/bashrc
set -u

export WM_PROJECT_USER_DIR=/home/<USER>/OpenFOAM/bhimsingh-v1706
export FOAM_USER_APPBIN="$WM_PROJECT_USER_DIR/platforms/$WM_OPTIONS/bin"
export FOAM_USER_LIBBIN="$WM_PROJECT_USER_DIR/platforms/$WM_OPTIONS/lib"

export PATH="$FOAM_USER_APPBIN:$PATH"
export LD_LIBRARY_PATH="$FOAM_USER_LIBBIN:${LD_LIBRARY_PATH:-}"

hash -r
```

Confirm the principal environment variables:

```
echo "$WM_PROJECT_DIR"
echo "$WM_PROJECT_USER_DIR"
echo "$WM_OPTIONS"
g++ --version | head -1
```

Validated values:

```
WM_PROJECT_DIR=/home/<USER>/OpenFOAM/OpenFOAM-v1706
WM_PROJECT_USER_DIR=/home/<USER>/OpenFOAM/bhimsingh-v1706
WM_OPTIONS=linux64GccDPInt320pt
g++ 7.5.0
```

6. Confirm the hy2Foam executable and catalytic libraries

If you use the preserved compiled tree on a compatible environment, first confirm that the executable and shared libraries resolve correctly. If the platform differs, rebuild from source in the validated Ubuntu 20.04/GCC7 toolchain.

```
which hy2Foam
hy2Foam -help | head -30

ls -lh "$FOAM_USER_LIBBIN"/libhy2*CatalyticWall.so

ldd "$(which hy2Foam)" | grep 'not found' || echo "PASS: hy2Foam dependencies resolved"

for L in \
  "$FOAM_USER_LIBBIN/libhy2PartialCatalyticWall.so" \
  "$FOAM_USER_LIBBIN/libhy2FullyCatalyticWall.so" \
  "$FOAM_USER_LIBBIN/libhy2SuperCatalyticWall.so"
do
  echo "==== $L ====="
  ldd "$L" | grep 'not found' || echo "PASS"
done
```

Expected production libraries:

- libhy2PartialCatalyticWall.so
- libhy2FullyCatalyticWall.so
- libhy2SuperCatalyticWall.so

The obsolete monolithic libhy2CatalyticWall.so is not part of the validated production configuration.

7. Catalytic-wall validation cases

Validated case root:

```
/home/<USER>/OpenFOAM/bhimsingh-v1706/run/catalytic_validation
```

Case	Library / wall model	Key behavior
------	----------------------	--------------

nonCatalytic	libstrathFiniteVolume.so	Reference non-catalytic wall
partialCatalytic	libhy2PartialCatalyticWall.so	gamma = 0.01 for N/N2 and O/O2 reactive pairs
fullyCatalytic	libhy2FullyCatalyticWall.so	gamma = 1 for N/N2 and O/O2 reactive pairs
superCatalytic	libhy2SuperCatalyticWall.so	Wall mass fractions prescribed from shared equilibrium properties

7.1 Use the validated mesh

For each catalytic case, the validated production mesh is the backup_polyMesh_final_5degrees copy.

```
cd /home/<USER>/OpenFOAM/bhimsingh-v1706/run/catalytic_validation/<CASE>

rm -rf constant/polyMesh
cp -a constant/backup_polyMesh_final_5degrees constant/polyMesh

checkMesh > log.checkMesh.validation 2>&1
tail -n 25 log.checkMesh.validation
```

The validated mesh reported:

- max aspect ratio approximately 7.77
- max non-orthogonality approximately 8.88 degrees
- max skewness approximately 0.396
- Mesh OK.

Do not use the historical 1000_baseline mesh as a validation reference. That case exhibited severe mesh-quality problems including extreme skewness and failed mesh checks.

7.2 Standard short smoke settings

```
foamDictionary system/controlDict -entry startFrom -set startTime
foamDictionary system/controlDict -entry startTime -set 0
foamDictionary system/controlDict -entry endTime -set 1e-8
foamDictionary system/controlDict -entry writeInterval -set 1e-9
```

During validation, adaptive stepping used a maximum Courant target of approximately 0.01 and the cases completed in 34 solver iterations.

7.3 Isolate the correct library

Partial catalytic:

```
foamDictionary system/controlDict -entry libs \
-set '("libstrathFiniteVolume.so" "libhy2PartialCatalyticWall.so")'
```

Fully catalytic:

```
foamDictionary system/controlDict -entry libs \
-set '("libstrathFiniteVolume.so" "libhy2FullyCatalyticWall.so")'
```

Super catalytic:

```
foamDictionary system/controlDict -entry libs \
-set '("libstrathFiniteVolume.so" "libhy2SuperCatalyticWall.so")'
```

8. Expected smoke-test results

Run a case with:

```
hy2Foam > log.hy2Foam.validation 2>&1
echo "HY2FOAM_EXIT_CODE=$?"
```

Minimum PASS criteria:

- process exit code is 0
- the solution advances to Time = 1e-08
- the log terminates with End
- written times include 1e-09 through 1e-08
- no FOAM FATAL, segmentation fault, signal 8, NaN, or actual floating-point exception
- the normal startup line 'sigFpe : Enabling floating point exception trapping' is not a failure

Case	Validated outcome	Final short-transient wall heat flux
nonCatalytic	PASS	32512.5412682 W
partialCatalytic	PASS	32512.5412682 W
fullyCatalytic	PASS	32512.5412682 W
superCatalytic	PASS	32675.5276503 W

Important: the identical integrated heat flux printed for the non-, partial-, and fully-catalytic cases at only 1e-08 s does not imply that those wall models are inactive. Their heavy-species fields and written wall gradients are distinct. The interval is deliberately a runtime smoke test, not a converged heat-flux comparison.

8.1 Partial catalytic response

The partial-catalytic run changed all heavy-species solution files relative to the non-catalytic run, while the electron field remained unchanged. Reactive N2/O2/N/O fields wrote nonuniform wall gradients.

- Reactive probability: gamma = 0.01
- N2: sourceSpecies N, fluxSign -1
- N: sourceSpecies N, fluxSign +1
- O2: sourceSpecies O, fluxSign -1
- O: sourceSpecies O, fluxSign +1

8.2 Fully catalytic response

The fully-catalytic run also changed every heavy-species field relative to both non-catalytic and partial-catalytic cases, while e- remained identical. The reactive entries retain gamma = 1.

- N/N2 and O/O2 reactive gradients are substantially stronger than the gamma = 0.01 partial case.
- This provides direct evidence that the fully-catalytic boundary condition is active.

9. Super-catalytic equilibrium verification

The super-catalytic wall uses constant/superCatalyticWallProperties. In the validated AIR-11 example, the prescribed wall equilibrium mass fractions are:

Species	Y_wall
N2	0.78
O2	0.22
NO	0
N	0
O	0
N2+	0
O2+	0
NO+	0
N+	0
O+	0
e-	0

The validated final written wall values matched these targets exactly: maximum species error = 0.0 and maximum closure error $|\text{sum}(Y_{\text{wall}}) - 1| = 0.0$.

10. Troubleshooting and known pitfalls

Symptom	Interpretation / action
Modern host GCC fails to compile old source	Use Ubuntu 20.04 with GCC/G++ 7.5.0, matching the validated toolchain.
hy2Foam not found	Source OpenFOAM bashrc, set WM_PROJECT_USER_DIR, FOAM_USER_APPBIN, and PATH.
Custom library cannot be opened	Check FOAM_USER_LIBBIN and LD_LIBRARY_PATH; run ldd on the .so file.
controlDict references patch 'object'	Use the actual wall patch name 'wall' in the validated cases.
Warning: patchGroup 'wall' clashes with patch	Observed mesh metadata warning; nonfatal in the validated run.
SCEBD test says runtime diagnostic block not found	The regression script expects special diagnostic markers not emitted by the normal production source; this alone is not a physics failure.
checkMesh reports extreme skewness / failed checks	You are likely using a historical/non-production mesh. Restore backup_polyMesh_final_5degrees.
Only sigFpe startup line appears in error grep	Normal OpenFOAM startup message, not an actual FPE.

11. Reproducibility checklist

1. Download both release parts and SHA256SUMS.
2. Verify SHA-256 checksums before reconstruction.
3. Concatenate part-00 and part-01; run gzip -t.
4. Extract OpenFOAM-v1706, ThirdParty-v1706, and bhimsingh-v1706.
5. Use Ubuntu 20.04 + GCC/G++ 7.5.0 or an equivalent compatible container.
6. Source OpenFOAM-v1706/etc/bashrc and export the user bin/lib paths.
7. Confirm hy2Foam startup and shared-library dependencies.
8. Use backup_polyMesh_final_5degrees for the validated catalytic cases.
9. Isolate the matching custom library in controlDict.
10. Run the 0 to 1e-08 smoke interval and confirm exit code 0, Time = 1e-08, and End.
11. For superCatalytic, verify the prescribed AIR-11 wall composition and closure.

Validated reference paths

```
/home/<USER>/OpenFOAM/OpenFOAM-v1706
/home/<USER>/OpenFOAM/ThirdParty-v1706
/home/<USER>/OpenFOAM/bhimsingh-v1706
/home/<USER>/OpenFOAM/bhimsingh-v1706/hyStrath
/home/<USER>/OpenFOAM/bhimsingh-v1706/run/catalytic_validation
```

Redistribution note: before making the repository public, review and preserve the license and attribution requirements of OpenFOAM, hyStrath, and bundled ThirdParty components.

Release integrity record

Release tag: v1.0.0-validated

part-00 SHA-256: 7129bf024e57f47a3c32dec29dd4d37eccc19a11e2fa5251663ca00bc404f81c

part-01 SHA-256: 2860ba19c2de9ca1f61cd4e549503d98b53ecd50a8a08a81fcb0de89d6a9bfe4